

NGUYEN NGOC TIN

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EDUCATION

Saigon University (SGU) — Ho Chi Minh City, VietnamMar 2021 – Jun 2026

Engineer in Information Technology — Major: Information Systems

Language Proficiency: TOEIC 625 / 990 (Jun 2026)

SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, Java (Core)
- **Backend and Web Frameworks:** Django REST Framework, FastAPI, Next.js (App Router), React, RESTful APIs, WebSockets, TailwindCSS
- **Cloud and AI Engineering:** AWS (Lambda, SAM, Bedrock, DynamoDB, S3), LLM Integration (Gemini, Bedrock, Ollama), Agentic Workflows, MCP
- **Databases and Architecture:** PostgreSQL, Redis, DynamoDB (Single-Table Design), SQLite, Clean Architecture, Event-Driven Architecture
- **DevOps and Tools:** Docker, Git, GitHub Actions, Linux

EXPERIENCE

FCAJ — Cloud Application Development InternMar 2026 – May 2026

Ho Chi Minh City, Vietnam

- Architected and deployed an event-driven serverless backend prototype utilizing Python and AWS Lambda, decoupling core business logic and optimizing asynchronous event processing.
- Constructed standardized AWS SAM templates and architectural documentation, establishing reproducible local emulation and streamlining deployment workflows for the engineering team.
- Configured cloud monitoring and structured logging pipelines across serverless components, ensuring environment consistency and accelerating error diagnosis during testing.

PROJECTS

Bếp Dì 6 – Online Ordering Platform (Zalo Mini App) | Django REST, React, PostgreSQL, RedisAug 2026 – Present

GitHub | Case Study

- Architected an end-to-end Food and Beverage online ordering system integrating a Zalo Mini App frontend (React 18, ZMP SDK) with a centralized Django REST API and PostgreSQL database.
- Engineered immutable order snapshotting and UUIDv4 idempotency keys via transaction.atomic() and Redis, eliminating duplicate checkouts and securing financial data integrity.
- Built a dynamic delivery fee engine using GPS Haversine distance calculations and integrated automated VietQR payment generation to streamline checkout flows.

Lexi – AI-Powered English Speaking Tutor | Next.js, TypeScript, Python, AWSMar 2026 – May 2026

GitHub | Demo Video

- Architected a cloud-native serverless backend using AWS SAM, API Gateway, Lambda, Cognito, DynamoDB, and S3, enforcing Clean Architecture across Domain, Application, and Infrastructure layers.
- Engineered a real-time voice streaming pipeline via API Gateway WebSockets, orchestrating Amazon Transcribe, Bedrock, and Polly for low-latency conversational feedback.
- Designed a DynamoDB Single-Table schema with composite partition/sort keys and GSIs, supporting 6+ distinct access patterns for user sessions, flashcards, and dialogue history with single-digit millisecond latency.

Bilingual Movie-Based English Learning Platform | Next.js, TypeScript, FastAPI, Python, SQLiteMar 2026 – Present

GitHub | Demo Video

- Architected a full-stack language learning web application using Next.js App Router and FastAPI, delivering synchronized bilingual movie playback and contextual vocabulary acquisition.
- Implemented subtitle parsing and binary-search timestamp alignment algorithms with SQLite indexing, enabling instantaneous contextual vocabulary lookups during video streaming.
- Integrated Google Gemini API with structured prompt engineering to generate contextual grammar breakdowns, phonetic explanations, and collocations directly from movie dialogues.